

FP-WDDQN: An improved deep reinforcement learning algorithm for adaptive traffic signal control

Xiao Zhang

Jiangsu Key Laboratory of Big Data Security & Intelligent Processing
Nanjing University of Posts and Telecommunications
Nanjing, Jiangsu, China
1221045523@njupt.edu.cn

Xiaolong Xu

School of Computer Science
Nanjing University of Posts and Telecommunications
Nanjing, Jiangsu, China
xuxl@njupt.edu.cn

Abstract—Current adaptive traffic signal control methods based on centralized deep reinforcement learning are not applicable in large-scale adaptive traffic control environment. The scalability problem is overcome by assigning global control to each local RL agent through multi-intelligence reinforcement learning, but the environment now becomes partially visible and non-stationarity from the perspective of each local agent due to limited communication between agents. In this paper, we propose a multi-agent framework called Forgetful Priority Weighed Double Deep Q-learning (FP-WDDQN). We firstly extend Weighted Double Deep Q-Learning(WDDQN) to the multi agent domain, so as to reduce the error caused by the algorithm's underestimation of the target network and get more accurate Q-value. Then we propose a new algorithm based on Forgetful Experience Mechanism(FEM) and Priority Experience Replay Mechanism(PERM). This mechanism makes WDDQN choose experience based on Temporal Difference (TD) error and Identify the importance of experience with FEM. It makes WDDQN preferentially select experience with high TD-error in the process of sampling and training its network based on FEM and TD-error, which improved the model training efficiency and stabilized the learning process. In this paper, we construct an urban traffic network with seven intersections in the simulation platform called Simulation of Urban MObility(SUMO), and the proposed FP-WDDQN is compared against Independent Double Deep Q-learning(IDDQN), WDDQN and Multi-agent Advantage Actor Critic (MA2C) under the condition of simulating the traffic dynamics of peak hours. Experiment shows FP-WDDQN has better performance than other algorithms in vehicle speed, intersection delay and intersection waiting queue length.

Keywords—Deep reinforcement learning, Multi-agent technology, Intelligent traffic light control, Forgetful Experience Mechanism, Prioritized Experience Replay.

I. INTRODUCTION

One of the ways to alleviate traffic congestion is to realize intelligent control of traffic lights. In the current society, most traffic lights still take the fixed time set in advance [1,2,3] for traffic control. The disadvantages of timed traffic control are obvious. Because the parameters of the signal phase and time design scheme are fixed and cannot be changed according to

the real-time traffic volume changes, it usually happens at traffic intersections that there are no cars in the green light while there are many waiting cars in the red light, which makes the conditions at intersection low efficiency. Some recent studies have proposed traffic light control rules based on heuristic algorithms based on real traffic data [3,4,5,6], but these rules are still difficult to dynamically adjust according to real-time flow in the face of large-scale traffic network environment. In order to realize real-time control of traffic lights, scholars have applied reinforcement learning technology [7,8,9,10] to intelligent traffic control. Unlike traditional model-driven approaches, reinforcement learning does not rely on heuristic assumptions and equations. Instead, it directly uses parametric models to learn optimal controls based on its experience interacting with complex transportation systems. However, traditional reinforcement learning faces two difficult challenges: how to represent the environment and how to model the relationship between the environment and the policy. In order to solve these two problems, recent studies have applied deep reinforcement learning techniques [11,12,13,14,15] to traffic light control problems. The use of deep neural networks greatly improves the scalability of single-agent reinforcement learning and enhances its learning ability. Although neural networks improve the scalability of reinforcement learning, it is still not feasible to train a centralized deep reinforcement learning agent for large-scale traffic network signal control problems.

Our main contributions can be summarized as follows.

1) We extend the weighted double Q learning to the multi-agent framework, which can estimate Q values more accurately and improve the performance of Q learning.

2) In order to solve the non-stationarity question in multi-agent framework, we design a new Forgetful Priority Experience Replay Mechanism based on Forgetful Experience Mechanism and Priority Experience Replay Mechanism, and then we introduce this mechanism into the multi-agent environment based on WDDQN

. The algorithm can preferentially select the experience with high error and strong timeliness in the process of training.

3) We constructed a road network at intersection 7 in the simulation platform called sumo, and the proposed algorithm is tested in the simulation platform. It is proved that FP-WDDQN can help reduce the instability of the environment and improve the convergence of the algorithm in the limited communication environment.

The remainder of this article is structured as follows. Section II investigates the current intelligent traffic light control methods based on deep reinforcement learning algorithm and analyzes the advantages and disadvantages of each method. Section III gives the details of the proposed Multi-agent deep reinforcement learning method FP-WDDQN. The model and algorithm are evaluated in Section IV. Section V concludes this paper.

II. RELATED WORK

A. Traffic Light Control based on Reinforcement learning

Since previous methods cannot handle dynamic multiway traffic, more studies attempt to use reinforcement learning algorithms to solve the traffic light control problem. These algorithms usually take the traffic on the road as the state and the operation of the signal light as the action. These methods generally show better performance than fixed time and traffic response control methods. Wen et al. [16] et al. proposed a single intersection signal light control method based on model-free SARSA algorithm. This method takes into account the actual traffic conditions, adopts stochastic control mechanism, and designs a specific state space by dividing the number of vehicles into sparse discrete values. Abudlhai et al. [17] proposed a single intersection signal control method based on Q-learning algorithm, which takes the queue length of vehicles as the state and the total delay between two actions as the reward function. Camponogara et al. [18] extended the Q-learning to the multi-agent framework by using distributed Q-learning technology and assigning independent Q values to each individual agent. This method can achieve good traffic light control in the traffic network with multiple intersections, but the state design of this method is discrete value, so it cannot be applied to large state space. With the increase of road flow and the expansion of the number of intersections, the model is difficult to show good performance.

B. Traffic light control based on deep reinforcement learning

The standard reinforcement learning algorithm is difficult to calculate the value function or policy function of all states in the high-dimensional state space. In contrast, it is easier to calculate the value function or policy function by using the f function approximation of deep learning. In this function approximation, deep neural network training is used to learn the optimal value function. At present, many neural network structures, such as recurrent neural network (RNN) and convolutional neural network (CNN), have been applied in large-scale state space and shown good performance. By using the function approximation

of Deep Learning to calculate the value function or policy function, recent studies [19,20] proposed the Deep q-learning algorithm using continuous state representation, that is, by Learning a Q function to map the state and behavior to obtain rewards. These studies verified that in the simplified traffic environment, Deep Q-learning has excellent fitting ability. These studies are different in state representation, including queue length [21], average delay [22], etc. Meanwhile, they are also different in reward design, including average travel time [23], average delay, etc. These algorithms usually only consider a single factor in the setting of state value and reward value. However, the state space of traffic intersection is often complex, and it is difficult to comprehensively include all factors of traffic flow only by queue length or average delay. Shanmugasundaram et al. [24] proposed a solution to the traffic signal control of different traffic flows at isolated intersections using double deep Q network, which used target network, update delay and empirical revisit mechanism. The model has the phenomenon of optimistic estimation of Q network, and it is difficult to show good performance with the increase of road flow.

C. Traffic light control based on multi-agent deep reinforcement learning

Genders et al. [25] applied Independent Deep Q-Learning, with the partial observability addressed with transfer planning, but the states were infeasible and the simulated traffic environments were oversimplified in that study. Foerster et al. [26] proposes using a multi-agent variant of importance sampling to naturally decay obsolete data. However, it didn't take into account the effect of time on experience. Shi et al. [27] proposed a multi-intersection control mechanism based on DQN-RNN, adopted a transfer planning technology suitable for smaller intersection sets, and used max-plus coordination algorithm to connect the learning results to larger intersection set. Zhang et al. [28] extended the proposed weighted double estimator to the multi-agent domain and propose a multi-agent DRL framework, named weighted double deep Q-network (WDDQN). Zheng et al. [29] extend the weighted double estimator to the multi-agent domain and propose a multi-agent DRL framework called weighted double depth Q network. It introduced a loose reward network and scheduling replay strategy to achieve effective cooperation in the multi-agent domain. Chu et al. [30] recently expanded Independent Deep Q-Learning to independent A2C for multi-agent TSC. The stability problem is addressed with two methods, fingerprints of neighbor intersections and a spatial discount factor.

Most existing methods are inspired by Independent Q-learning which extended the single-agent Reinforcement Learning approach directly to traffic environments. These researches are all faced with a problem that agent strategy is non-stationary in multi-agent environments, these algorithms cannot learn good cooperation strategies in complex road network environments.

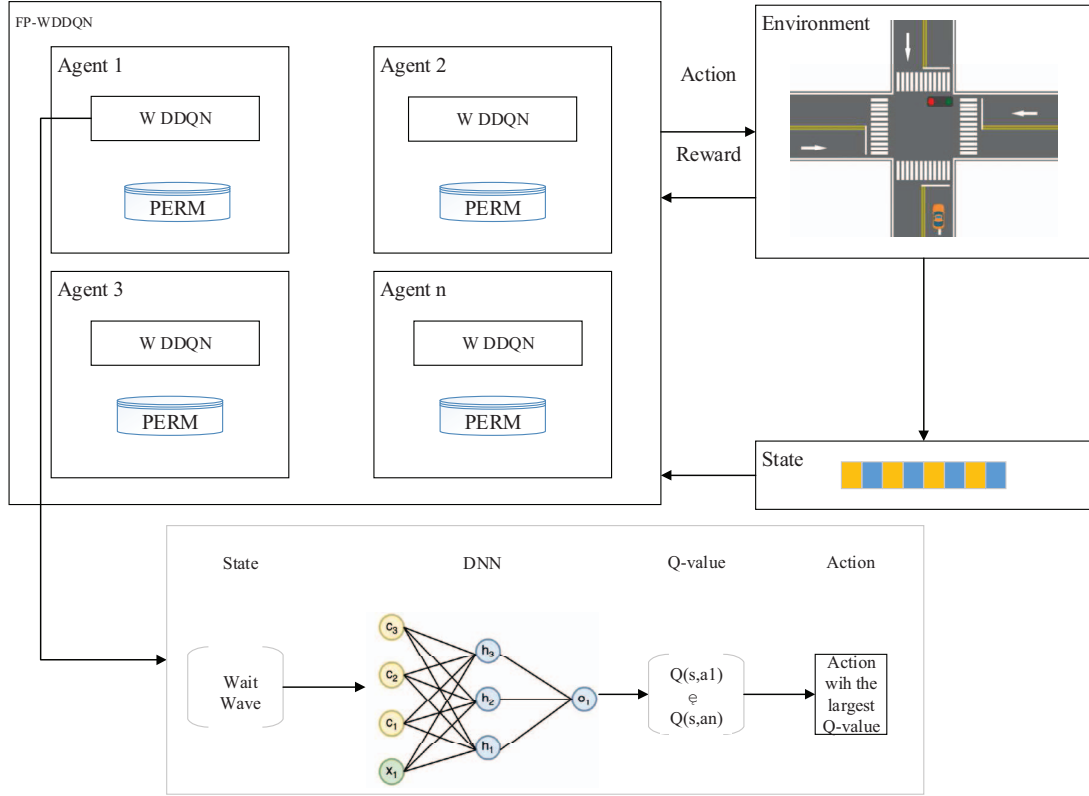


Fig. 1. Intelligent traffic light control based on FP-WDDQN

III. ALGORITHM

A. The framework of FP-WDDQN

The proposed FP-WDDQN improves the intelligent light control method based on deep reinforcement learning in the following three aspects:

1) Combined deep Q network and deep double Q network in the way of weight [28], so as to reduce the error of target network estimation.

2) The Prioritized Experience Replay is introduced to ensure that important experiences are preferentially selected in the process of agent training and improve the efficiency of agent training.

3) The observability of each local agent is improved by transferring the discounted state value of the agent in the adjacency node, and the algorithm is extended to multi-agent application scenarios to achieve multi-agent training.

The intelligent traffic light control process based on FP-WDDQN is shown in Figure 1:

Step 1: The agent obtains the local traffic state s in the current stage through the simulation platform;

Step 2: The agent takes the current traffic state s as the input of Q-neural network;

Step 3: The agent randomly selects the next stage action a with probability ϵ , and selects the maximum value of the neural network output sequence with probability $1-\epsilon$ as the next stage action a .

Step 4: The simulation platform executes the action a provided by the agent to make the traffic state enter the next stage s' and return the immediate reward value r ;

Step 5: Generate the experience tuple $\langle s, a, s', r, e_t=1 \rangle$ based on the above steps, and store the experience tuple into the experience pool.

Step 6: Select several experience tuples from the experience pool by the preferential Prioritized Experience Replay Mechanism every once in a while to train the Q-neural network.

B. Weight Double-DQN

Aiming at the problem that Double-DQN underestimates agent Q value, this section firstly combines deep Q network and deep Double Q network in the way of weight, so as to reduce the error of target network estimation and obtain more accurate Q value estimation. Secondly, the Prioritized Experience Replay is introduced to ensure that important experience is preferentially selected in the process of agent training.

Double-DQN contains two networks: the evaluation network and the target network. The evaluation network generates an action with a maximum Q value, and the target network evaluates the Q value for that maximum action. In this section, the evaluation network and the target network are used to calculate the Q value of action A , and the new target value is generated based on the weight. The calculation formula is defined as follows:

$$y^{DDQN} = r + \gamma Q(s', \argmax_a Q(s', a, \theta), \theta^-) \quad (1)$$

WDDQN uses a dynamic heuristic value β , which depends on a constant c , to balance between the overestimation of the single estimator and the underestimation of the double estimator during the iterative Q-value update process:

$$y^{WDDQN} = r + \gamma[\beta Q(s', a^*; \theta) + (1 - \beta)Q(s', a^m; \theta^-)] \quad (2)$$

Where $Q(s', a^*; \theta)$ is the maximum output Q value of the target network by state s' , $Q(s', a^m; \theta^-)$ is the minimum Q value of the evaluation network output in state s' , β is defined as the weight, the calculation formula is defined as follows:

$$\beta = \frac{|Q(s', a^*; \theta^-) - Q(s', a^m; \theta^-)|}{c + |Q(s', a^*; \theta^-) - Q(s', a^m; \theta^-)|} \quad (3)$$

where c is a constant used to calculate the weight action β . a^* is the action with the maximum Q value obtained by evaluating the network, and action a^m is the action with the minimum Q value obtained by evaluating the network. The calculation methods can be defined as follows:

$$a^* = \argmax_a Q(s', a; \theta) \quad (4)$$

$$a^m = \argmin_a Q(s', a; \theta) \quad (5)$$

C. Forgetful Priority Experience Replay Mechanism

Independent Q-Learning relies on Experience Replay Mechanism (ERM) to achieve great success in the early single agent framework. While in the multi-agent framework, the environment is unstable. As the agent is updated, the environment also changes, which leads to a change in the importance of the experience stored earlier. ERM's random sampling no longer reflects the true importance of experience. In this section, we design a priority experience replay mechanism based on FEM and TD-error.

We store a time factor e_t for each experience when it is generated, which will decay with the change of training times. It indicates the importance of this experience for training. the experience stored in PERM is defined by $\langle s, a, s', r, e_t \rangle$. We initialize e_t to be 1, it decreases by a decay rate d_e when an episode is terminated.

We combine TD-error and time factor e_t as the criteria to evaluate the importance of experience. Take the experience i as an example, experience i is defined by $\langle s_i, a_i, s'_i, r_i, e_i \rangle$, its priority is calculated as follows:

$$\delta_i = r_i + \gamma[\beta Q(s'_i, a_i^*; \theta) + (1 - \beta)Q(s'_i, a_i^m; \theta^-)] - Q_w(s_i, a_i) \quad (6)$$

The loss function of FP-WDDQN is calculated as follows:

$$Loss = \frac{1}{B} \sum_{x \in B} (\delta_i * e_i)^2 \quad (7)$$

where B means the number of experiences to be selected in a training session.

Now we have defined the calculation method of experience priority value, However, we need to learn not only high-value lessons, but also low-value lessons with the aim to explore more environments and avoid overfitting of network. So we take a probabilistic approach to extracting experiences to make sure that all experiences have a chance to be extracted. The probability of each experience is defined as follows.

$$P_i = \frac{\delta_i + \mu}{\sum (\delta_i + \mu)} \quad (8)$$

where μ is a small constant, to ensure that experience with zero TD-error still has the probability of being extracted.

D. FP-WDDQN based on multi-agent framework

In this section, we combine WDDQN and Forgetful Priority Experience Replay Mechanism, design a new algorithm called FP-WDDQN.

In order to ensure that with the migration of time, each agent pays more attention to the use of the existing experience for training, so as to gradually reach a stationary state, in this section, the ϵ -greedy strategy used to explore the new experience in the agent is modified. In the past, every time an agent interacts with the environment, it will randomly select an action with a fixed ϵ probability and add this experience into the experience pool. This practice makes it impossible for the agent to dynamically adjust the proportion of exploration and utilization with the change of time. This section changes ϵ from a fixed value to a dynamic value that linearly decreases with time, ensuring that the agent focuses more on exploring the environment when it lacks experience in the early stage, and more on training the Q network based on existing experience in the later stage. The calculation of ϵ is defined as follows:

$$\epsilon = \max(\epsilon_{min}, \epsilon_{init} * (1 - \frac{t}{T})) \quad (9)$$

Based on the above improved methods, the process of FP-WDDQN is shown in Algorithm 1. We take an agent i in FP-WDDQN as an example, agent i uses ϵ -greedy strategy to balance the exploration and exploitation. whenever agent i interact with the environment, it will put an experience into a tuple, which is defined as $\langle s, a, s', r, e_t = 1 \rangle$; When agent i choose to train, it will extract B experiences from the experience pool based on PERM to train the neural network.

Algorithm 1: FP-WDDQN for Agent i

-
1. Initialize Parameters $M_i, N, \gamma, \epsilon, d_e, B, \mu, c, \epsilon_{min}, \epsilon_{init}$.
 2. Initialize evaluation network and target network with parameters $\theta_i^- = \theta_i$.
 3. **For** episode = 1 to N **do**
 4. **For** $t = 1$ to not terminal **do**
 5. get current intersection state s_t and select an action a_t by θ_i with ϵ -greedy strategy.
 6. observe its reward r_t and next observation s'_t , after all agents execute their actions.
 7. Store experience $\langle s_t, a_t, s'_t, r_t, e_t = 1 \rangle$ in M_i .
 8. **If** $step > M_i$ **then**
 9. drop the experience with smallest Loss, which can get by Eq.6 and Eq.7.
 10. update evaluate network θ_i with a probabilistically sampled mini-batch from M_i .
 11. update the target networks: $\theta_i^- = \theta_i$.
 12. **end for**
 13. Update e_t for all experience in M_i by $e_t \leftarrow d_e * e_t$.
 14. **end for**
-

E. Agent design

In this section we design the state, action and reward principle of the agent.

(1) State

This section is set for an intersection. For each lane $i \in L_i$, i at this intersection, $wait$ measures the cumulative delay of the first vehicle, while $wave$ measures the total number of approaching vehicles along each incoming lane.

(2) Action

In this experiment, we designed four phases (signal combination of traffic lights) for each intersection based on reality: north-south straight(NS), east-west straight(EW), north-south left(NSL), east-west left(EWL). the actions of agent correspond to four different phases, each time according to the selected action into the corresponding phase. Agent's action controlled by ϵ - greedy strategy, agent in the probability of ϵ environment to explore, namely randomly selected an action; $1-\epsilon$ to assess the probability of using the current network makes the Q value of the largest movements.

(3) Q network structure

All networks in the experience use the same architecture and hyper-parameters. We perform evaluations using a DDQN architecture as a basis for the algorithm. The Q networks of IDDQN, WDDQN, and FP-WDDQN consist of 2 fully connected layers and an output neuron for each action. The Structure is shown in Fig. 2.

(4) Reward

The setting of the reward value of the agent has an important influence on its learning ability, so it is necessary to select the index which can correctly reflect the current traffic section when setting the reward value. In this experiment, the reward value of the agent is defined as the weighted sum of the following factors in order to ensure the objective and true reflection of the agent's learning ability to traffic conditions:

$$r_{i,t} = - \sum_{j \in \epsilon, l \in L_{ji}} (queue_{t+\Delta t}[l] + a * wait_{t+\Delta t}[l]) \quad (10)$$

where a is a tradeoff coefficient, and $queue$ is the queue length along each incoming lane. $wait$ measures the cumulative delay of the first vehicle. This reward is post-decision so both $queue$ and $wait$ are measured at time $t + \Delta t$, it is directly correlated to state and action, and emphasizes both traffic congestion and trip delay.

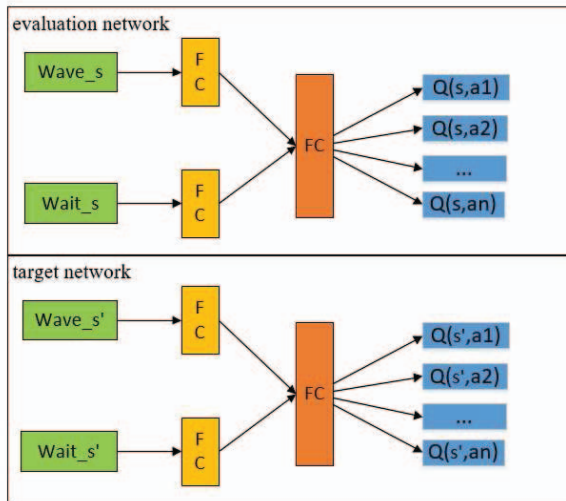


Fig. 2. Q Network Structure

(5) the size of step

In this paper, Δt is defined as the period of interaction between the agent and the traffic environment, which means that the agent will change the current policy each time after the environment has simulated the duration of Δt . If the Δt design is too long, the agent will find it difficult to adapt; if the Δt design is too short, the agent will not be able to deliver the policy on time considering the communication delay and computation overhead issues. Moreover, the strategy of changing traffic lights quickly will also bring about traffic safety problems.

IV. EXPERIMENTS AND ANALYSIS

A. Experimental environment

In this paper, the SUMO (Simulation of Urban Mobility) Simulation platform is used as the experimental environment. SUMO software provides flexible API for road network design, traffic volume simulation, and traffic light control. Specifically, the simulation of real road and traffic flow can be realized on the platform, and the traffic movement can be controlled according to the given traffic light policy (obtained by the intelligent traffic light algorithm). The environment used for this experiment is a traffic network with seven intersections. In addition to crossover 4, each intersection is connected with four sections of 150 meters long, and each road has two lead-in lanes and two lead-out lanes. The traffic lights in this part of the experiment include two stages: (1) the east-west green light (east-west green light, south-north upward red light), (2) the east-west red light (east-west red light, south-north upward green light). When there is a green light in one direction, there is a red light in the other direction. In addition, the green light is followed by a yellow light for 3 seconds and then changes to a red light. In order to test the robustness and optimality of the agent in this traffic scenario, random and time-varying traffic flows are designed to simulate the traffic during rush hours. Specifically, with the introduction of 1000 cars per hour, three sets of additional traffic flows are designed. F_1 consists of three sets of start-end pairs (X1-X2), (X1-X3), (X4-X2), F_2 consists of three sets of start-end pairs (X4-X3), (X4-X6), (X4-X7), F_3 consists of two start-end pairs (X7-X1), (X7-X4). During the experiment, every 15 minutes, F_1 and F_2 alternately became the main traffic flow, and the other two groups acted as the branch traffic flow. The main traffic flow will generate an integer multiple of 400 vehicles/hour "flow" in the specified area, while the secondary traffic flow will generate an integer multiple of 200 vehicles/hour "flow" in the specified area. The road condition is shown in Fig. 3.

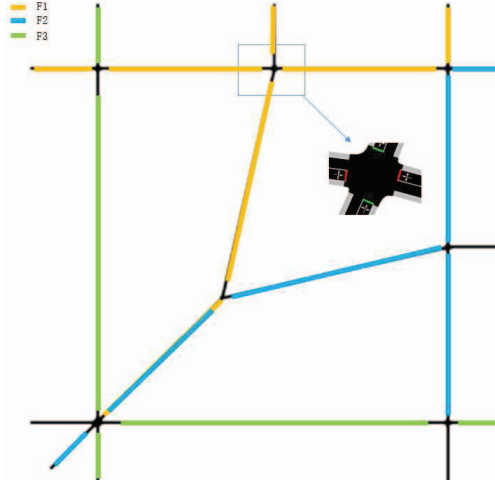


Fig. 3. Seven intersection road network

Some specific parameters of traffic environment are shown in TABLE I :

TABLE I. ENVIRONMENTAL PARAMETERS

Environmental parameters	Setting
Extra traffic flow	1000 vehicles
Simulation step Δt	5 seconds
yellow light time t_y	2 seconds

B. Agent parameter Settings

This experiment trained the agent algorithm for more than 0.5 M steps, which is about 700 episodes for a given event step $T=720$ steps. Agent related parameters in this experiment are set as follows: the size of agent experience pool is set to 1000; In each training, 20 experiences are selected according to the Prioritized Experience Replay Mechanism. Refer to Table II for other parameters.

TABLE II. AGENT PARAMETERS

Agent parameters	Value
PERM size	1000
constant c	0.1
mini-batch B	20
learning rate	0.0001
ϵ -greedy ϵ_{init}	0.8
ϵ -greedy ϵ_{min}	0.2
Future impact factor γ	0.99
Decay rate d_e	0.97

C. The experiment indicators

In this section, the algorithm is evaluated for more than 10 episodes and the final average is taken. Results The model will be comprehensively evaluated based on the following indicators.

Average queue length: the average queue length over a period of time, where the queue length at time t is the sum of all vehicles in the approaching lane. A shorter queue means fewer cars waiting in all lanes.

Average delay: the average delay changing with time. The delay at time t is the sum of D of all approaching lanes. Lower delays mean higher speeds in all lanes.

Average Vehicle speed: The average speed at which a vehicle passes through an intersection. A higher average speed indicates a more open road, while a lower average speed indicates a more congested road.

Total number of vehicles passing through the intersection: the total number of vehicles continuously passing through the intersection within the time interval Δt after adopting the strategy.

D. Benchmark algorithm

IDDQN, WDDQN, and MA2C were selected as the control group in this experiment.

IDDQN: IDDQN regards other agents as part of the environment directly, that is, each agent is solving a single agent task. Obviously, since there are agents in the environment, the environment is unsteady, which cannot guarantee convergence, and the agents are easy to be trapped in wireless exploration.

WDDQN: By utilizing the weighted double estimator and the deep neural network, WDDQN can not only reduce the bias effectively but also be extended to scenarios with raw visual inputs. To achieve efficient cooperation in the multi-agent domain, it introduces the lenient reward network and the scheduled replay strategy.

MA2C: A2C is one of the most popular reinforcement learning algorithms based on the idea of policy improvement. It is different from DDQN based on value improvement. By constructing a policy-based neural network, the agent can directly output the policy in the interaction with the environment. MA2C expanded Independent Deep Q-Learning to independent A2C for multi-agent TSC. The stability problem is addressed with two methods, fingerprints of neighbor intersections and a spatial discount factor.

Based on the above experimental parameter Settings, the following experimental results are obtained.

From Fig. 4 we can see, IDDQN can't converge within 500000 steps. WDDQN and MA2C gradually converge after experiencing turbulence. While FP-DDQN can converge earlier and maintain a high reward value.

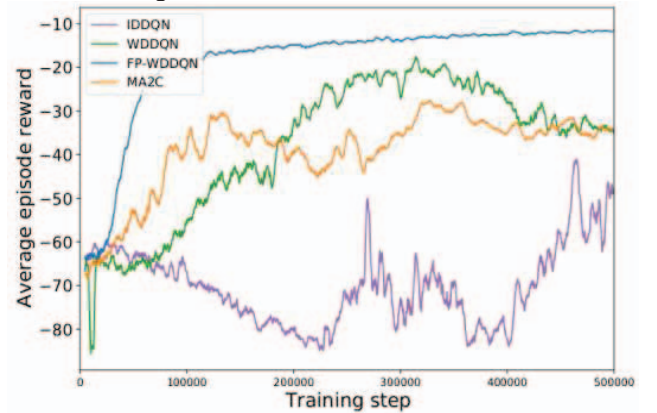


Fig. 4. Training step

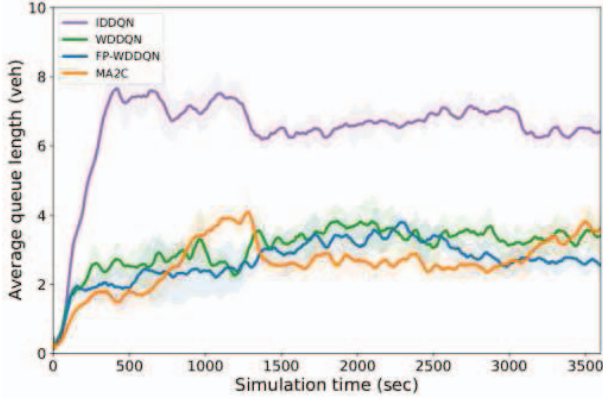


Fig. 5. Avg_queue_length

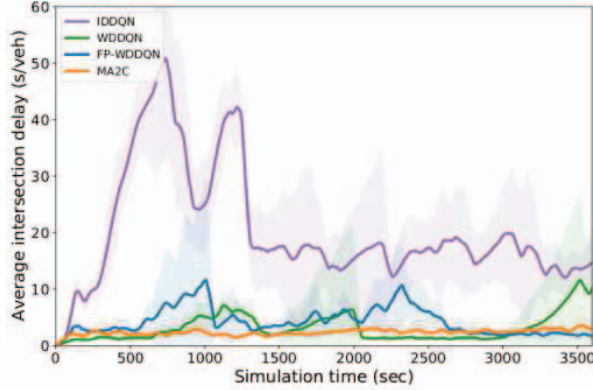


Fig. 6. Avg_intersection_delay

Fig. 5 shows the average queue length and the shaded figure shows the standard deviation over the entire evaluation period. It can be seen that IDDQN is difficult to alleviate the upcoming traffic congestion through policies. In contrast, WDDQN and FP-WDDQN strategies can optimize the queue length by maximizing the traffic at each step, and their performance is better.

Fig. 6 plots the average intersection delay of the network during the simulation time. As expected, the delay of the IDDQN policy increases monotonically because it cannot recover from congestion. WDDQN can gradually reduce the delay in the late simulation time, but it can be seen that it has the displayed optimization waiting time, and the performance is lower than the FP-WDDQN, because of the high variance of the learning strategy.

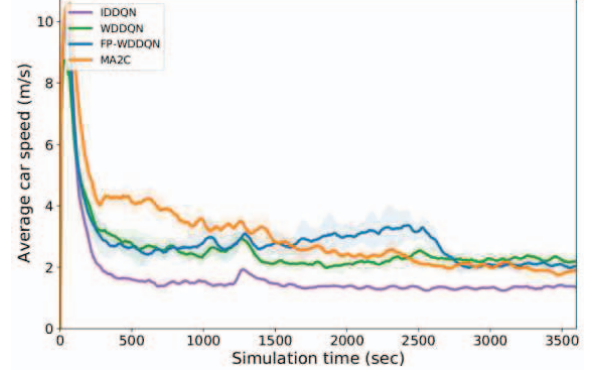


Fig. 7. Avg_car_speed

Fig. 7 plots the average vehicle speed of the network during the simulation time. It can be seen that both WDDQN and IDDQN fail to show good performance. MA2C based on policy update can maintain a slower delay growth rate, but it is generally worse than FP-WDDQN. FP-WDDQN based on value update shows a good performance in general, although the performance of FP-WDDQN suffers from oscillation in the second half of simulation time. The reason for the oscillation is the problem that the value update algorithm can hardly avoid, namely the instability of the environment.

TABLE III lists the key evaluation indexes based on IDDQN, WDDQN, MA2C and FP-WDDQN in intelligent traffic light control. FP-WDDQN outperforms other algorithms. In almost all indexes. It can be seen that the algorithm has better robustness, optimality, efficiency and security.

TABLE III. PERFORMANCE IN SEVEN SIMULATION TRAFFIC NETWORK

Indexes	Algorithm			
	IDDQN	WDDQN	MA2C	FP-WDDQN
Avg. queue length	6.54	2.74	3.09	2.70
Avg. intersection delay	20.59	4.28	3.38	3.18
Avg. car speed	1.67	3.04	3.60	3.84

V. CONCLUSION

In this paper, FP-WDDQN is proposed to solve the problems existing in intelligent traffic light control. Compared with the traditional DDQN, the proposed algorithm is estimated by both target network and evaluation network Status - The value of the action. This method can reduce the error of Q estimation of target network. We then extend the WDDQN to a multi-agent framework. In order to solve the problem of unstable environment brought by multi-agent framework, a priority

experience replay mechanism based on forgotten experience mechanism and TD-error is designed in this paper. Specifically, every kind of menstruation. When a piece of experience is generated, the value e_t is added to indicate the importance of experience, and it decays at the rate of d_e . When the agent selects the experience for training, TD-error decides which experience to choose preferentially. The significance of the experience is identified by its experience e_t . In this way, the agent can preferentially select the experience with large errors.

For the early experience, the agent weakens its importance and makes the agent pay more attention to the new generation. This kind of experience improves the convergence ability of the algorithm. In this paper, an urban traffic network composed of 7 intersections is constructed on the simulation platform SUMO to simulate the traffic dynamics during peak hours. We compare FP-WDDQN with popular IDDQN, WDDQN, and MA2C. Experiment shows that FP-WDDQN has good robustness, optimality and expansibility.

ACKNOWLEDGMENT

We would like to thank the editors and reviewers for their comments and for helping us improve the quality of this paper.

REFERENCES

- [1] H. Rakha, F. Dion, "comparison of delay estimates at under-saturated and over-saturated pre-timed signalized intersections," *Transportation Research Part B: Methodological*, vol. 38, no. 2, pp. 99-122, 2004.
- [2] N. Wu, D. Li, and Y. Xi, "Distributed Integrated Control of a Mixed Traffic Network With Urban and Freeway Networks," *IEEE Transactions on Control Systems Technology*, vol. 30, no. 1, pp. 57-70, 2022.
- [3] D. Pekaslan, C. Wagner, and J. M. Garibaldi, "ADONiS—Adaptive Online Nonsingleton Fuzzy Logic Systems," *IEEE Transactions on Fuzzy Systems*, vol. 28, no. 10, pp. 2302-2312, Oct. 2020.
- [4] M. Ali, A. Durdu, S. A. Celtek, and A. Yilmaz, "An Adaptive Method for Traffic Signal Control Based on Fuzzy Logic With Webster and Modified Webster Formula Using SUMO Traffic Simulator," *IEEE Access*, vol. 9, pp. 102985-102997, 2021.
- [5] H. Ceylan, M. G. Bell, "Traffic signal timing optimisation based on genetic algorithm approach, including drivers routing," *Transportation Research Part B: Methodological*, vol. 38, no. 4, pp. 329-342, 2004.
- [6] S. Darmoul, S. Elkosantini, and A. Louati, "Multi-agent immune networks to control interrupted flow at signalized intersections," *Transportation Research Part C: Emerging Technologies*, vol. 82, pp. 290-313, 2017.
- [7] S. Zhan, "Research on Application of Deep Reinforcement Learning in Traffic Signal Control," in *2021 6th International Conference on Frontiers of Signal Processing (ICFSP)*, Paris, France, Apr. 15, 2021, pp. 17-21.
- [8] G. Borrageiro, N. Firoozye, and P. Barucca, "The Recurrent Reinforcement Learning Crypto Agent," *IEEE Access*, vol. 10, pp. 38590-38599, 2022.
- [9] M. Volodymyr, K. Koray, and S. David, "Human-level control through deep reinforcement learning," *Nature*, vol. 518, no. 7540, pp. 529-533, 2015.
- [10] Y. Zhu, X. Yin, R. Li, and C. Chen, "Extracting Decision Tree from Trained Deep Reinforcement Learning in Traffic Signal Control," in *2021 International Conference on Cyber-Physical Social Intelligence (ICCSI)*, Beijing, China, Dec.18-20, 2021, pp. 1-7.
- [11] M. Greguri, M. Vuji, and C. Alexopoulos, "Application of Deep Reinforcement Learning in Traffic Signal Control: An Overview and Impact of Open Traffic Data," *Applied Sciences*, vol. 10, no. 11, pp. 4011, 2020.
- [12] L. Lyu, Y. Shen, and S. Zhang, "The Advance of Reinforcement Learning and Deep Reinforcement Learning," in *2022 IEEE International Conference on Electrical Engineering*, Banda Aceh, Indonesia, Sep. 27-28, 2022, pp. 644-648.
- [13] S. Do, C. Lee, "Multi-agent Reinforcement Learning in a Large Scale Environment via Supervisory Network and Curriculum Learning," in *2021 21st International Conference on Control, Automation and Systems (ICCAS)*, Beijing, China, Nov. 05-07, 2021, pp. 207-210.
- [14] W. Z. Tareq, M. F. Amasyali, "Merging Reinforcement Learning and Inverse Reinforcement Learning via Auxiliary Reward System," in *2022 International Conference on Artificial Intelligence in Information and Communication (ICAIIIC)*, Guam, USA, Dec.12-15, 2022, pp. 292-297.
- [15] D. Xie, C. Chen, and Z. Wang, "IEDQN: Information Exchange DQN with a Centralized Coordinator for Traffic Signal Control," in *2020 International Joint Conference on Neural Networks (IJCNN)*, Glasgow, United Kingdom, Jul.19-24,2020, pp. 1-8.
- [16] K. Wen, S. Qu, and Y. Zhang, "A stochastic adaptive control model for isolated intersections," in *2007 IEEE International Conference on Robotics and Biomimetics (ROBIO)*, Sanya, China, Dec.15-18, 2007, pp. 2256-2260.
- [17] B. Abdulhai, R. Pringle, and G. J. Karakoulas, "Reinforcement Learning for True Adaptive Traffic Signal Control," *Journal of Transportation Engineering*, vol.129, no.3, pp. 278-285, 2003.
- [18] E. Camponogara, W. Kraus, "Distributed Learning Agents in Urban Traffic Control," in *Portuguese Conference on Artificial Intelligence Springer Berlin Heidelberg*, Springer, Berlin, Heidelberg, Dec.04, 2003, pp. 324-335.
- [19] T. Chu, S. Qu, and J. Wang, "Large-scale multi-agent reinforcement learning using image-based state representation," in *2016 IEEE 55th Conference on Decision and Control (CDC)*, Las Vegas, NV, USA, Dec. 12-14, 2016, pp. 7592-7597.
- [20] A. Salaymeh, L. Schwiebert, and S. Remias, "Multi-Agent Reinforcement Learning for Optimizing Traffic Signal Timing," in *8th International Conference on Artificial Intelligence and Applications (AIAP 2021)*, Vienna, Austria, Oct.08-29, 2021, pp. 1140-1150.
- [21] P. Mannion, J. Duggan, and E. Howley, "An Experimental Review of Reinforcement Learning Algorithms for Adaptive Traffic Signal Control," *Springer International Publishing*, 2016, pp. 47-66.
- [22] W. Genders, S. Razavi, (2016). Using a Deep Reinforcement Learning Agent for Traffic Signal Control, [Online]. Available: <http://arxiv.org/abs/1611.01142>.
- [23] E. Pol, A. Oliehoek, (2016). Coordinated Deep Reinforcement Learners for Traffic Light Control, [Online]. Available: <https://dare.uva.nl>.
- [24] P. Shanmugasundaram, A. Sinha, "Intelligent Traffic Control using Double Deep Q Networks for time-varying Traffic Flows," in *2021 8th International Conference on Signal Processing and Integrated Networks (SPIN)*, Noida, India, Aug. 26-27,2021, pp. 64-69.
- [25] W. Genders, S. Razavi, (2016). Using a deep reinforcement learning agent for traffic signal control, [Online]. Available: <http://arxiv.org/abs/1611>.
- [26] Foerster, Jakob, "Stabilising experience replay for deep multi-agent reinforcement learning," in *International conference on machine learning*. PMLR, 2017, pp. 1146-1155.
- [27] Shi, C. Feng, "Deep Recurrent Q-learning Method for Area Traffic Coordination Control." on *Journal of Advances in Mathematics and Computer Science*, 2018, pp. 1-11.
- [28] Z. Zhang, Z. Pan, and M. Kochenderfer, "Weighted Double Q-learning," in *Twenty-Sixth International Joint Conference on Artificial Intelligence*, Melbourne, Australia, Aug.19-25, 2017, pp. 3455-3461.
- [29] Zheng, Yan, "Weighted double deep multi agent reinforcement learning in stochastic cooperative environments," in *15th Pacific Rim International Conference on Artificial Intelligence*, Nanjing, China, Aug.28-31, 2018, pp. 421-429.
- [30] Chu, Wang, and Li, "Multi-agent deep reinforcement learning for large-scale traffic signal control," in *IEEE Transactions on Intelligent Transportation Systems*, 2019, pp. 1086-1095.